

INTRODUCTION TO BUILT ENVIRONMENT

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VARIOUS ELEMENTS OF BUILT ENVIRONMENT

ARE :

1. HOME
2. OFFICES
3. COMMERCIAL BUILDINGS
4. PARKS AND RECREATION CENTERS

HOME :

Home is a primary need for human life. Having a beautiful home is a dream for many. A successful person is recognized by his home. Due to the large demand for homes, there is a promising market for real estate. This indicates the importance of home for human life.

Purpose Of House :

- Keep your Family
- Home is necessary for proper health
- House is a place of strong security
- If one is tired due to his work or other activities, he runs for his home
for relief from stress
- House is a place where one can enjoy complete freedom. He lives as he
likes and does what he wants

- Home is the place where one can equip maximum comforts like air conditioning, bathing, dining, bed, and other comforts to a maximum level
- Home is the best place for peace of mind
- Home is also indicative of status

OFFICES

The main purpose of an office environment is to support its occupants in performing their jobs. Work spaces in an office are typically used for conventional office activities such as reading, writing and computer work. There are nine generic types of work space, each supporting different activities

In addition to individual cubicles, one can find meeting rooms, lounges, and spaces for support activities, such as photocopying and filing. Some offices also have a kitchen area where workers can make their lunches.

Commercial building

A commercial building is a building that is used for commercial use. Types can include office buildings, warehouses, or retail (i.e. convenience stores, 'big box' stores, shopping malls, etc.)

Office Building: to serve the staffs for working.

Warehouse: to put the stock of logistic and the products you have made.

Retail: create the stores to sell the products like clothes, cosmetic, etc.

PARKS AND RECREATION CENTERS

Parks and recreation have three values that make them essential services to communities:

1. Economic value
2. Health and Environmental benefits
3. Social importance

Economic value

Parks improve the local tax base and increase property values. It is proven that private property values increase the value of privately owned land the closer such land is to parks. This increase in private property value due to the proximity to parks increases property tax revenues and improves local economies.

A per review of 25 studies investigating parks and open space contributed positively to the property values of surrounding properties found that 20 of the 25 studies found that property values were higher.

Health and Environmental benefits

Parks are the places that people go to get healthy and stay fit. Parks and recreation programs and services contribute to the health of children, youth, adults, and seniors.

Social importance

Parks provide gathering places for families and social groups, as well as for individuals of all ages and economic status, regardless of their ability to pay for access.

TRANSPORTATION

Transport (commonly used in the U.K.), or transportation (used in the U.S.), is the movement of humans, animals and goods from one location to another. In other words, the action of transport is defined as a particular movement of an organism or thing from a point A (a place in space) to a point B. Modes of transport include air, land (rail and road), water, cable, pipeline and space.

The field can be divided into infrastructure, vehicles and operations. Transport enables trade between people, which is essential for the development of civilizations.

Modes of Transportation

The Following are common modes of transportation.

1. Road Transportation or Roadways
2. Rail Transportation or Railways
3. Air Transportation or Airways
4. Water Transportation or Waterways

Apart from the above types of transportation, the following types of transportation are required in special cases:

- Conveyers
- Pipelines
- Ropeways

Road Transportation or Roadways

The branch of transportation engineering that deals with the planning, design, construction, and maintenance of roadway facilities is termed Road Engineering or Highway Engineering.

Among all types of the transportation road system is an only mode which could give maximum flexibility of service to reach from one place to the desired location.

Various classes of vehicles such as a car, bus, truck, two-wheeler, etc. may be permitted to make use of the roads. Apart from the road vehicles, pedestrians also make use of the facilities of the road system. Road transport mode has the maximum flexibility for travel with reference to choice of the route, direction, time and speed of travel.

Road transportation is the only mode that caters to the movement of passengers and goods independently right from the place of

origin up to the destination of any trip along with the land.

In other words, it is possible to provide a door to door service only by road transport.

Roads are of various kinds and may be classified on the following basis:

- Usages During Any Season
- Types of Pavement
- Width and Quality

Usages During Any Season

On this basis Road may be classified as

- i. All Weather Roads
- ii. Fair Weather Roads

Types of Pavement

On this basis Road may be classified as

- 1. Surfaced Road
- 2. Unsurfaced Road

Width and Quality

On this basis Road may be classified as

1. Expressway
2. National highways (NH)
3. State highways (SH)
4. Major District Road (MDR)
5. Other District Road (ODR)
6. Village roads (VR)

Rail Transportation or Railways

Rail transport (also known as train transport) is a means of transferring passengers and goods on wheeled vehicles running on rails, which are located on tracks. In contrast to road transport, where vehicles run on a prepared flat surface, rail vehicles are directionally guided by the tracks on which they run.

Rail transportation is classified on the basis of distance

between rails as –

1. Broad Gauge

2. Meter Gauge

3. Narrow Gauge

Air Transportation or Airways

Air transport is the fastest means of movement from one place to the other. It has reduced distances by minimizing the travel time.

It is very essential for a vast country like India, where distances are large, Air transport in India made a beginning in 1911 when airmail operation commenced over a little distance of 10 km between Allahabad and Naini.

Air travel can be separated into two general classifications:

- National/domestic and
- International flights.

Flights from one point to another within the same country are called domestic flights.

Flights from a point in one country to a point within a different country are known as international flights.

Water Transportation or Waterways

Water transport is the process of moving people, goods, etc. by boat, ship over a sea, ocean, lake, river, etc.

It operates on a natural track and hence does not require huge capital investment in the construction and maintenance of its track except in case of canals.

The cost of operation of water transport is also very less. It has the largest carrying capacity and is most suitable for carrying bulky goods over long distances.

It is a fuel-efficient and eco-friendly mode of transport.

The water transport is of two types—

(a) Inland waterways, and

(b) Oceanic waterways.

Water Resources

Water resources are natural resources of water that are potentially useful as a source of water supply. 97% of the water on the Earth is salt water and only three percent is fresh water; slightly over two thirds of this is frozen in glaciers and polar ice caps. The remaining unfrozen freshwater is found mainly as groundwater, with only a small fraction present above ground or in the air.

Natural sources of fresh water include surface water, under river flow, groundwater and frozen water.

Sources of water are :

1. Surface water
2. Groundwater
3. Rainwater
4. Water System

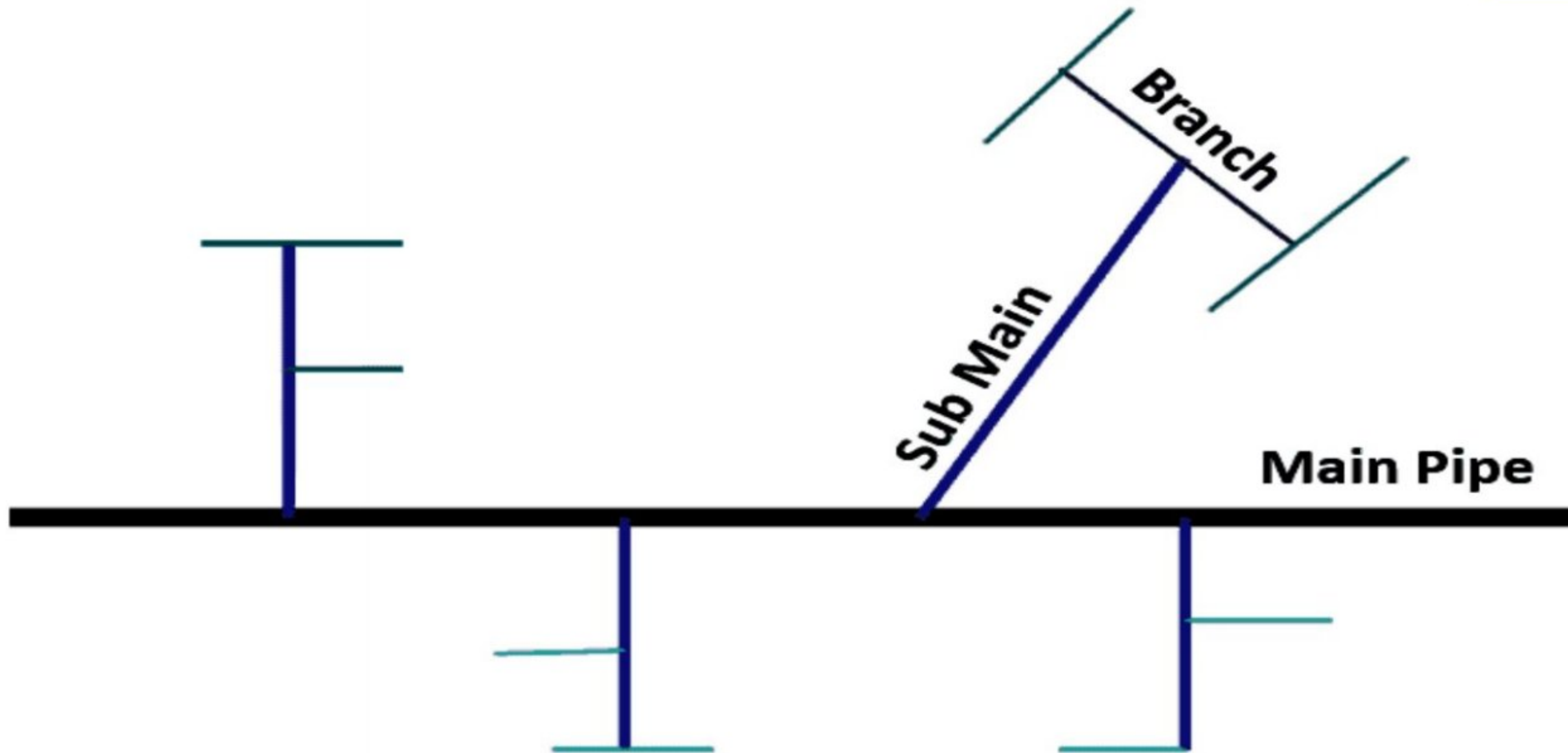
Types of Water Distribution System

The following are four major types of water distribution system

1. Dead-end or Tree Distribution system
2. Gridiron Distribution System
3. Circular or ring Distribution System
4. Radial Distribution System

Dead-end or Tree Distribution System

In this types of water distribution system, one main pipeline runs through the centre of the building, and the sub-mains branch lines off from both sides. The sub-main lines are then divided into several branch lines from which service connections are provided for particular houses.

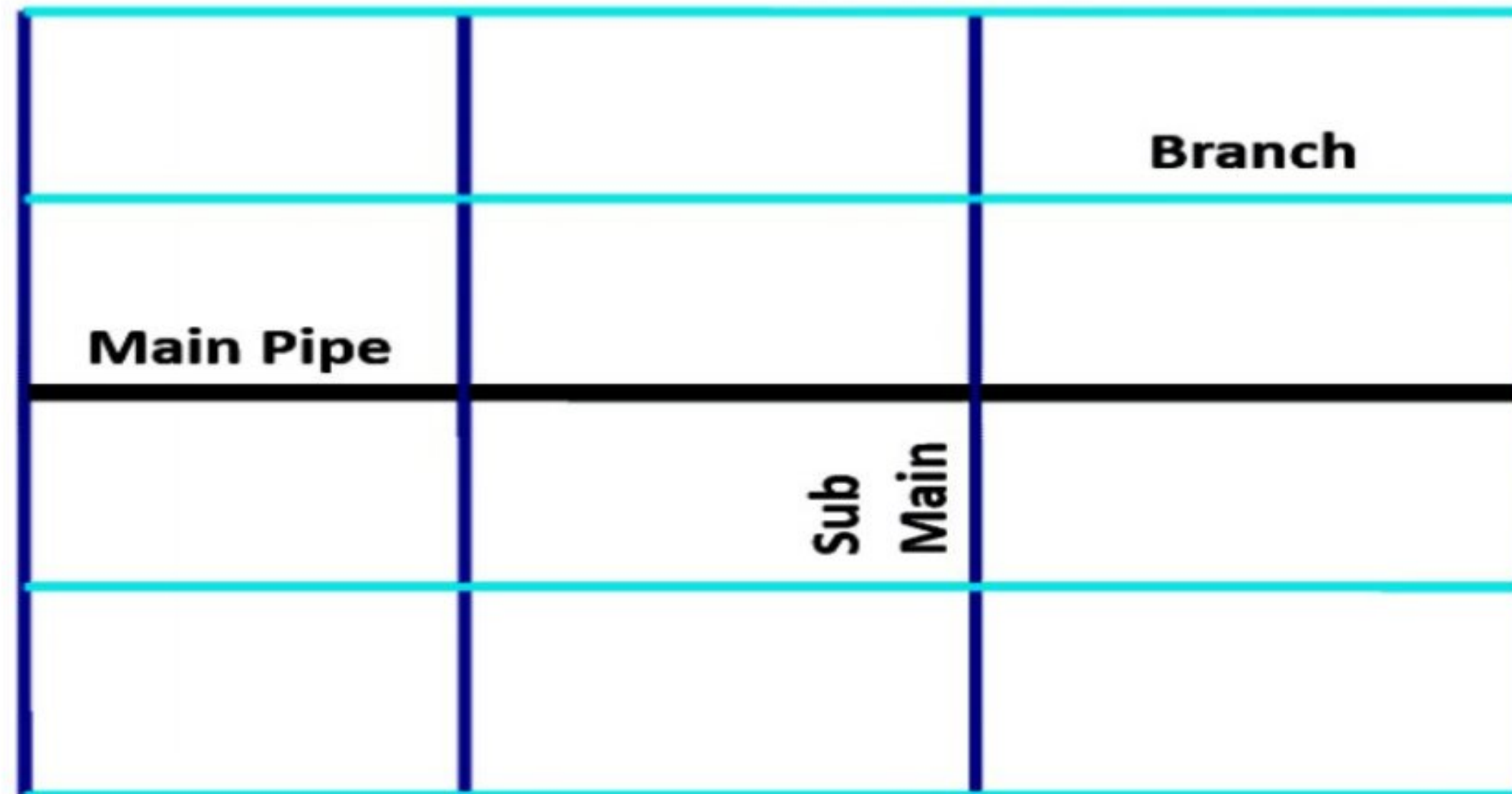


Gridiron Distribution System

In this, the main supply lines run through the center of the building, and sub mains branch off in perpendicular directions. The branch interconnects the sub-mains.

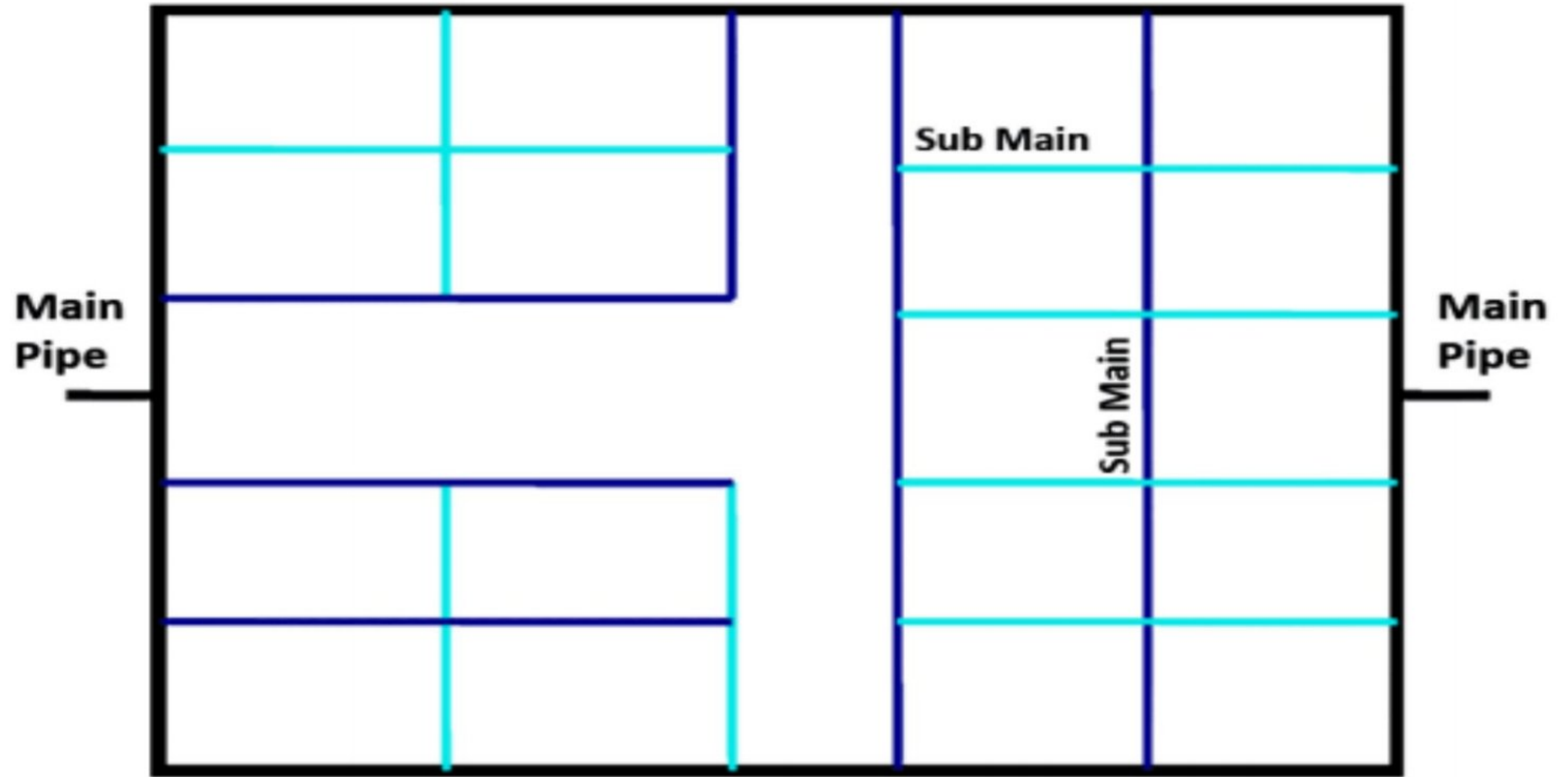
The distinguishing features of this system are that all the type of pipes are interconnected and there are no dead ends.

Water can reach at the given point of withdrawal from many directions, which allows more flexible operation, particularly when repairs are required



Circular or Ring Distribution System

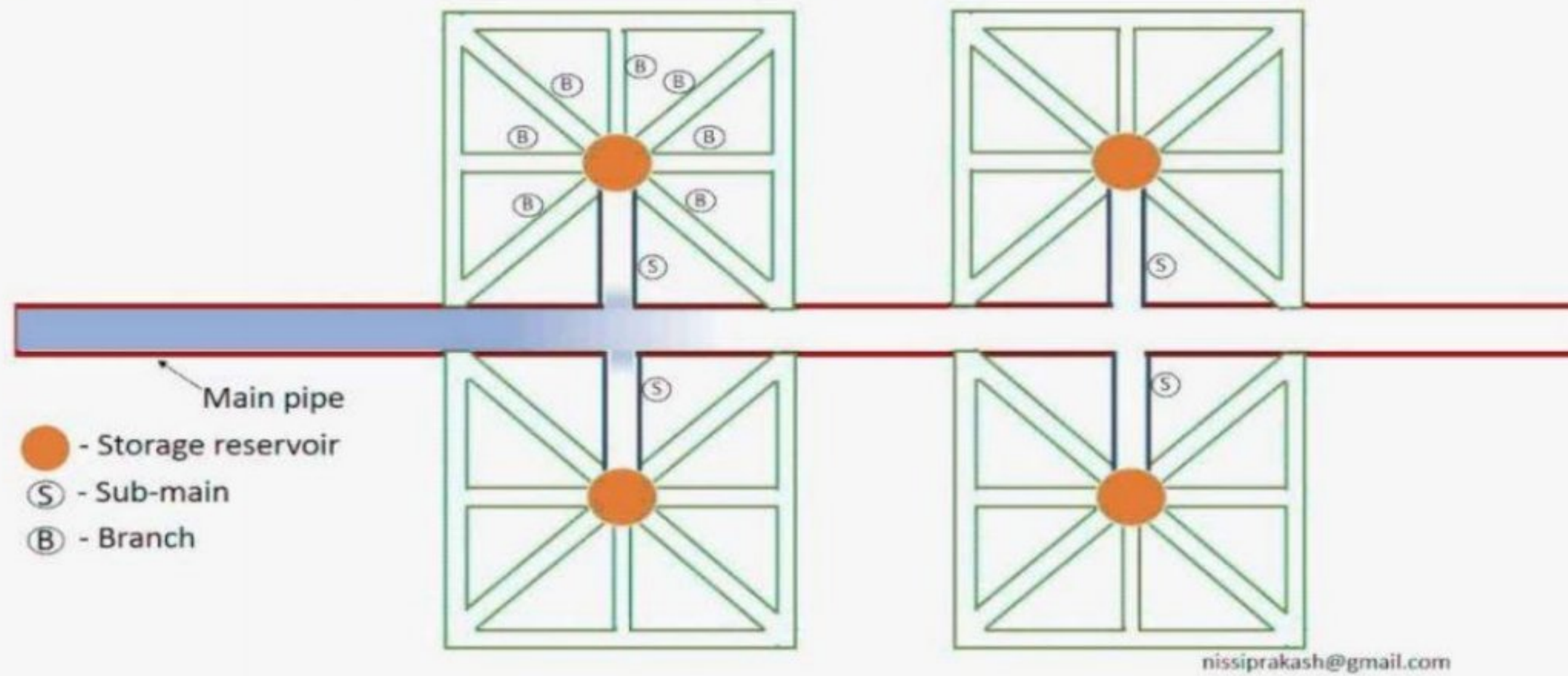
In this type of water distribution system, the supply mains forms a ring around the area. The branch pipes are connected cross-wise to the mains and also to each other. This system is most reliable for several buildings located in the same area planned locations.



Radial Distribution System:

In this types of water distribution system, the whole buildings are divided into several distribution areas. Each building has a centrally located elevated reservoir from where distribution pipes run radially towards the periphery of the distribution areas. This system gives swift service, without much loss of head.

RADIAL SYSTEM



Infrastructure

Infrastructure is the set of fundamental facilities and systems that support the sustainable functionality of households. Infrastructure is composed of public and private physical structures such as roads, railways, bridges, tunnels, water supply,sewers,and telecommunications (including Internet connectivity and broadband access).

Buildings

Buildings serve several societal needs – primarily as shelter from weather, security, living space, privacy, to store belongings, and to comfortably live and work. A building as a shelter represents a physical division of the human habitat (a place of comfort and safety) and the outside (a place that at times may be harsh and harmful).

Tunnel

A tunnel is an underground passageway, dug through the surrounding soil/earth/rock and enclosed except for entrance and exit, commonly at each end. A pipeline is not a tunnel, though some recent tunnels have used immersed tube construction techniques rather than traditional tunnel boring methods.

Dam

Dam, structure built across a stream, a river. Dams are built to provide water for human consumption, for irrigating arid and semiarid lands, or for use in industrial processes. They are used to increase the amount of water available for generating hydroelectric power, to reduce peak discharge of floodwater created by large storms or heavy snowmelt, or to increase the depth of water in a river in order to improve navigation and allow ships to travel more easily. Dams can also provide a lake for recreational activities such as swimming, boating, and fishing

Canal

It is an artificial channel constructed on the ground to carry water to the field either from a reservoir, tank or river.

There are three main sources from which we take water:

1. Reservoir.
2. Tank.
3. River.

The water in canal is used for the following purposes:

- 1.To power house for power generation.
2. To tanks for water supply.
3. To feed for irrigation.

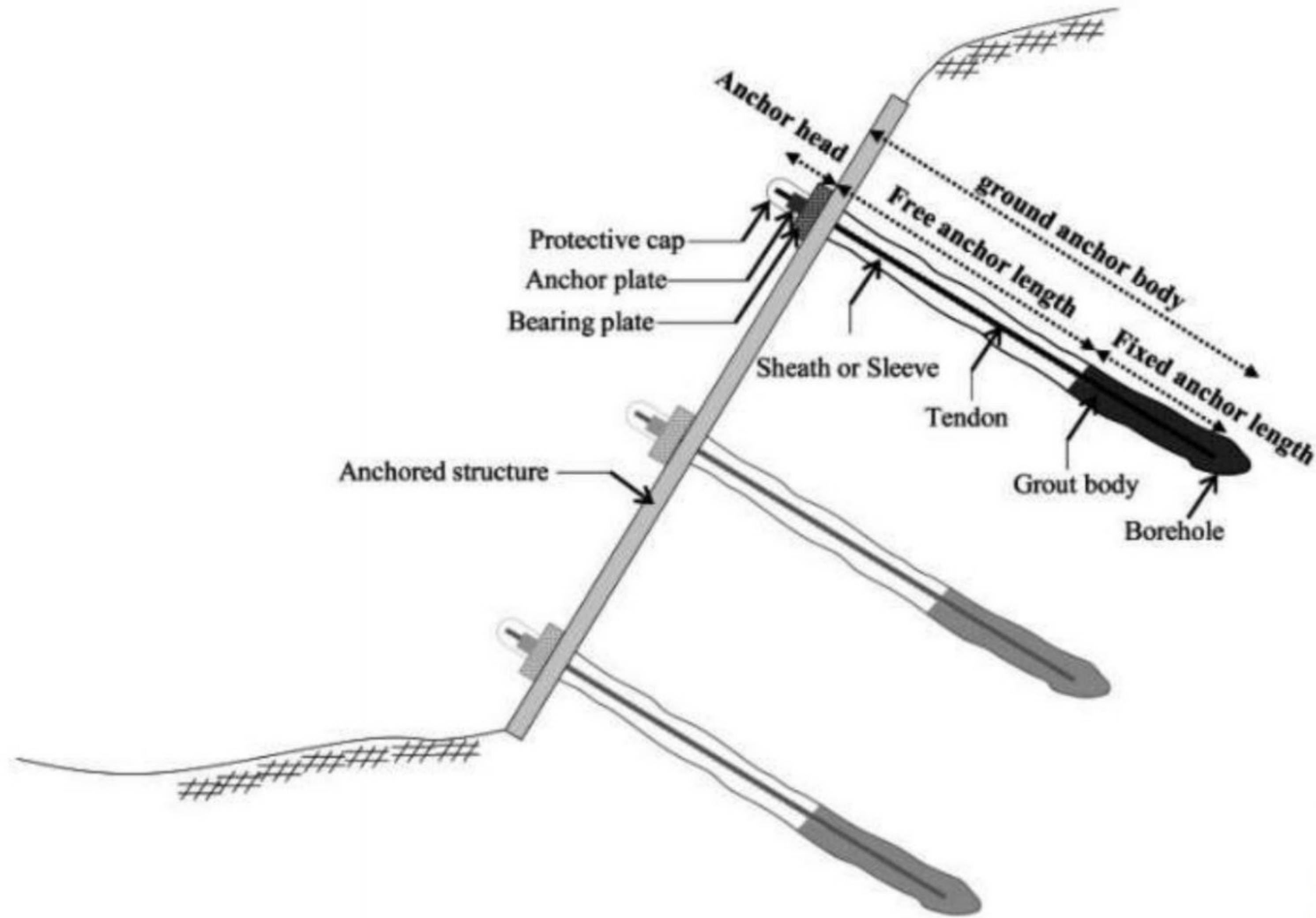
Sewerage system

A sewerage system, or wastewater collection system, is a network of pipes, pumping stations, and appurtenances that convey sewage from its points of origin to a point of treatment and disposal.

- Combined systems
- Separate systems and Alternative systems

Ground support system

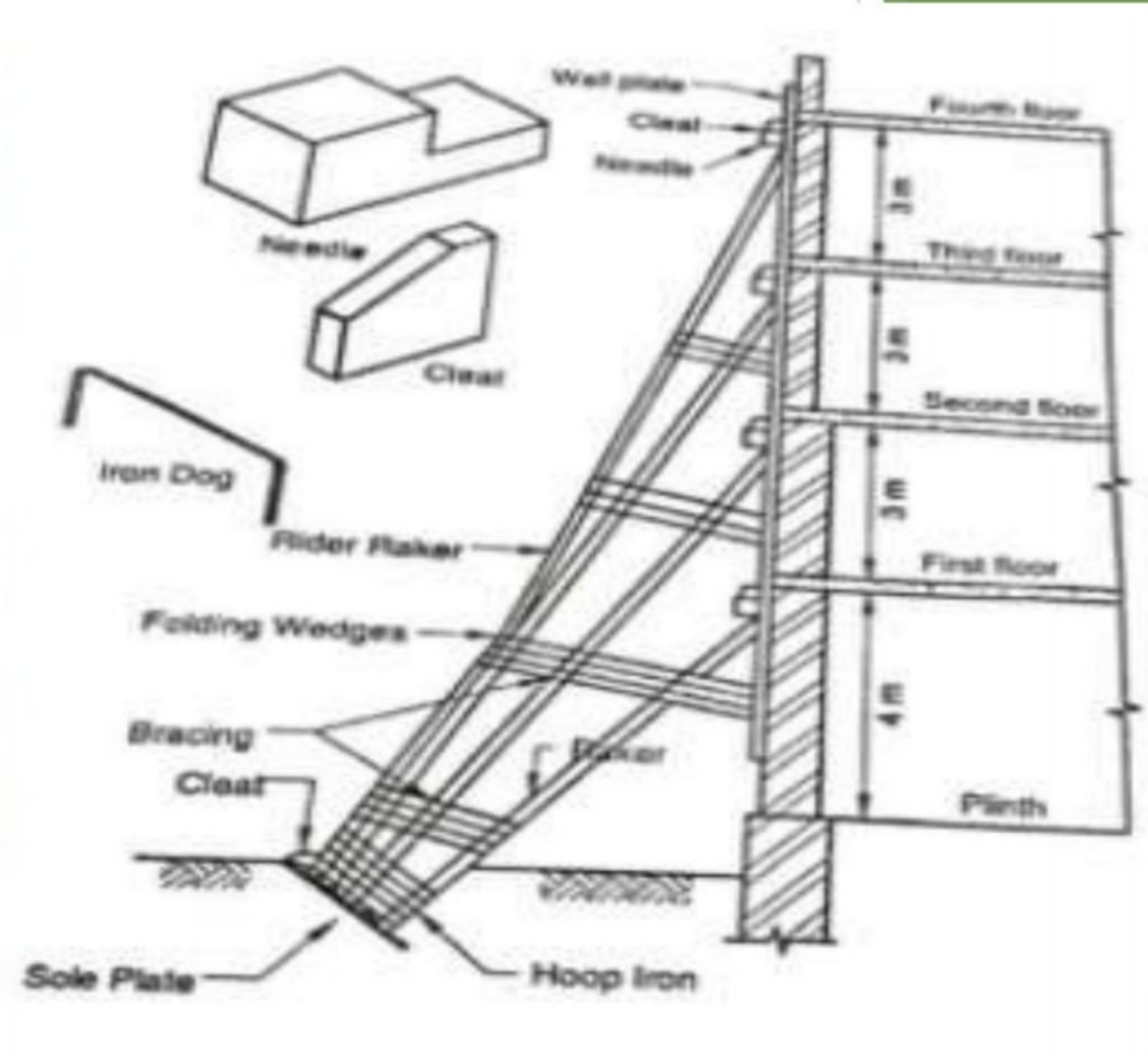
An anchor is a system used to transfer tensile loads to the ground. Depending on the design life of the anchor, it is classed as either a temporary or a permanent anchor. Permanent anchors usually have double corrosion protection. Ground anchors are always active, i.e. once injected and the grout has set, they are stressed up to the required load.



Shoring is used to support the beams and floors in a building while a column or wall is removed. In this situation vertical supports are used as a temporary replacement for the building columns or walls.

Types of Shoring :

1. Raking Shoring
2. Hydraulic Shoring



Underpinning is the process of strengthening and stabilizing the foundation of an existing building or other structure.

Underpinning may be necessary for a variety of reasons:

- The original foundation is simply not strong or stable enough, e.g. due to decay of wooden piles under the foundation.
- The usage of the structure has changed.
- The properties of the soil supporting the foundation may have changed (possibly through subsidence) or were mischaracterized during planning.

- The construction of nearby structures necessitates the excavation of soil supporting existing foundations.
- It is more economical, due to land price or otherwise, to work on the present structure's foundation than to build a new one.
- Underpinning is accomplished by extending the foundation in depth or in breadth so it either rests on a stronger soil stratum or distributes its load across a greater area. **Use of micropiles and jet grouting are common methods in underpinning.**